Homework 1
World Models in AI
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1. Why do we prefer AI models that learn an inner model of the world and how it works (from a geometric, temporal, dynamic, physics and semantics perspective)?

The main idea behind an AI model that learns an *inner model* of the world is that it can build an internal representation of its environment instead of simply reacting to its current observations. This representation can combine geometry, time, movement, physics and semantics, allowing the model to predict how the world may change and what consequences its actions might have. In very simple words, this is what people call "*intuition*" or "*what you would expect to happen*" in situations you have not encountered before. For example, if you train a classic Neural Network to classify images of **red dogs** and **blue cats**, having only seen images from these two categories, it would have a very hard time trying to classify an image of a red cat. A human, however, who understands the concept of color, would have a very easy time explaining that it is the same cat but in red. This very same *intuition* is also the one that allows humans to predict what would happen in different scenarios. For example, you have probably never seen a piano thrown out of the window of a house. But, while reading the previous sentence, you probably already formed an image in your mind and "*calculated*" how it would look smashing into the ground. This is mostly because we understand concepts like *gravity*, *breaking*, *weight* and already have our own *inner model* of the world.

This inner model is not to be confused with education or learning of certain topics. Notice how the piano example could still be imagined by anyone who has lived on this earth, even if they have never taken a physics class. This is also the most difficult part for building such an AI: giving it an inner model of the world is more than giving it a set of labeled data. It needs to develop a level of understanding we cannot really teach. It's much more than a set of rules.

Also, creating an inner model of the world can also introduce bias because the AI learns its understanding from the data and assumptions provided by humans. If the training data represents the world incompletely or disproportionately, the model may develop an inaccurate or biased representation of certain situations. Humans also implicitly decide which aspects of the world are important to represent, meaning that the design of the inner model itself can reflect human assumptions and biases.

2. Why is not enough to learn to predict the right class or action directly from patterns of data, without knowing "why" or having a model of how the world "works" and "is"?

It is not enough for an AI model to simply learn patterns and predict the correct class or action because the real world contains situations that may differ from its training data. A model based only on correlations may recognize that certain events usually occur together without understanding why they are related. An inner model of the world allows the AI to understand the relationships between objects and events and use them to make predictions in

new situations. This is especially important when the environment changes or when an object appears in an unusual context. Furthermore, understanding cause and effect allows the AI to predict the consequences of actions instead of simply relying on what happened in previous examples.

3. What is the advantage of having a model that considers many tasks and has many different interpretations of the world at the same time (it estimates geometry, movement of objects as well as their classes, together) as opposed to completely different models or different tasks?

One advantage of having one model that considers multiple aspects of the world at the same time is that these aspects are interconnected. For example, knowing that an object is a car is more useful when the model also knows its position, size, velocity and is trying to calculate its trajectory. Knowing which type of car and what its specifications are can only lead to a more accurate calculation. Another possible advantage is that a unified model could share information and representations instead of having several independent models performing similar computations. This can make the overall system more efficient. However, a unified model would be more complex and difficult to train, debug and maintain, while specialized models would be easier to replace or improve individually.

However, we also have to understand this discussion in our human context. People tend to specialise into certain fields, not because there is only so much information that our brains are able to accumulate, but because a specialised individual also offers more confidence in that specific field. For example, most people would prefer a surgeon that has been practicing medicine his entire file to one that was working as a car mechanic five years ago. Even if their surgeries have the same success rate in the present.

4. Why do we prefer intelligence that learns and discovers knowledge by itself and not just supervised by a teacher or known ground truth during training?

When it comes to learning, most people imagine similar things. For most of us, learning is strongly related to school and academic fields, but nowadays many online resources like tutorials or courses have also become popular. However, the truth remains that for most of humanity, "*learning*" itself is mostly supervised. We even use entirely different words for people who learn in unsupervised manners (inventor, discoverer, researcher). That's why it was natural that in the development of Artificial Intelligence, we would tend to model AI's *learning* based on our own.

However, as the field of AI develops, our expectations of it tend to grow larger. We have reached the point where we want to use AI to expand our field of knowledge, not just repeat stuff we already know. And this is one of the first ways in which a supervised approach reaches its limits, you can only "teach" what you already know. Just like it is very difficult for a student to learn more algebra than his teacher.

Moreover, unsupervised learning allows the model to discover patterns and links between data that humans may now have anticipated. This is particularly useful when dealing with a large amount of complex data, since manually defining all the relevant relationships

would be difficult and would also introduce unwanted bias into the training data. Instead of being restricted to the categories and relationships defined by a human teacher, the model can explore the structure of the data and identify new ways to interpret it. This, however, does not completely eliminate bias, since training data and learning objectives that we offer to the AI model still contain implicit human biases.